Eleanor Waiss

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Education

University of Colorado — Boulder, CO
Ph.D., Mathematics — *In progress*

Butler University — Indianapolis, IN
B.S., Mathematics and Actuarial Science — *May 2025*
Minor in Computer Science

Research Interests

Broadly speaking, I enjoy problems that have a computational element to them. I enjoy writing computer algorithms, testing ideas iteratively, and having concrete answers to play with. My current research interests lie in the intersection between dynamics, number theory, and computational mathematics. I am interested in complex dynamics and fractal geometry, computer visualizations of fractals, and interesting number-theoretic properties that arise in this study.

Publications and Preprints

Analysis of Algorithms for Moser's Problems on Sums of Consecutive Primes
INTEGERS Electronic Journal **26A**, #A24, 28 September 2026. Joint with J. Sorenson. (arXiv, Journal)

Geometric Limits of Sums of Iterated and Fixed Polynomials
submitted, awaiting review. Joint with A. Kapiamba and S. Kaschner. (preprint)

Presentations

Invited Talks

Developing Intentional Practices and Cultivating an Inclusive Environment for Trans, Non-Binary, and Genderqueer Students*, with D. Beh, L. McKittrick-Sweitzer, and E. Edwards. Inclusive Pedagogy Workshop, Butler University, Indianapolis, IN USA. March 2023.

Contributed Talks

Algorithms to Write Integers as Sums of Consecutive Primes. INTEGERS Conference on Number Theory, Athens, GA USA. May 2025. (slides)

An Atlas of Differential Topology: From Fundamentals to Stokes. Butler Undergraduate Research Conference 2025, Indianapolis, IN USA. April 2025. (slides)

A Gleeful Algorithm: Enumerating Sums of Consecutive Primes. Butler Undergraduate Research Conference 2025, Indianapolis, IN USA. April 2025. (slides)

Geometric Limits of Julia Sets of Sums of Iterates and Powers of Polynomials. Joint Mathematics Meetings, San Francisco, CA USA. January 2024. (slides)

Seminars

A Limited History of Complex Dynamics. Mathematics Colloquium Series, Butler University, Indianapolis, IN USA. April 2024. (slides)

Posters

A Gleeful Algorithm: Enumerating Sums of Consecutive Primes. BIGEAST Research Symposium, New York City, NY USA. March 2025.

Dabbling in Dobble!: An Exploration of Monomatching Games. MAA Indiana Fall Sectional, Terra Haute, IN USA. October 2022.

Teaching

Graduate Teaching Assistant August 2026 – Present
University of Colorado
- Calculus I – Fall 2026

Conferences and Workshops Attended

Midwest Dynamical Systems October 2026
University of Wisconsin-Madison, WI USA

Preliminary Arizona Winter School September 2025
(Virtual) University of Arizona, AZ USA

INTEGERS Conference on Number Theory May 2025
University of Georgia, GA USA

BIGEAST Research Symposium March 2025
New York City, NY USA

AMS Spring Central Sectional April 2024
University of Wisconsin-Milwaukee, WI USA

Joint Mathematics Meetings January 2024
San Francisco, CA USA

Dynamical Systems and their Applications September 2023
University of Miami, OH USA

MAA Indiana Fall Sectional October 2022
Rose-Hulman Institute of Technology, IN USA

Awards

Amos Carpenter Memorial Award May 2025
Butler University

Robert O. Whitesell Endowed Scholar May 2024
Butler University

Service

Senator, College of Arts and Sciences October 2026–Present
Graduate & Professional Student Government *University of Colorado*

Tutor Coordinator August 2024–May 2025
Department of Mathematical Sciences *Butler University*

President August 2022–May 2023
Actuarial Science Club *Butler University*

Industry

Actuarial Analyst January 2026–July 2026
West Bend Mutual Insurance *West Bend, WI*

Actuarial Intern May 2023–August 2023
OneAmerica Financial Partners *Indianapolis, IN*